

Function Transformations

DSAT Math Practice • 20 Questions • Mixed Difficulty

Total Questions: 20 | Multiple Choice: 17 | Grid-In (SPR): 3

Question 1 [Easy]

The function g is defined by $g(x) = f(x) + 5$. If the graph of $y = f(x)$ passes through the point $(2, 3)$, through which point does the graph of $y = g(x)$ pass?

- A) $(2, 8)$
 - B) $(7, 3)$
 - C) $(2, -2)$
 - D) $(7, 8)$
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Question 2 [Easy]

The graph of $y = f(x)$ is translated 4 units to the right to obtain the graph of $y = h(x)$. Which of the following defines $h(x)$?

- A) $h(x) = f(x + 4)$
 - B) $h(x) = f(x - 4)$
 - C) $h(x) = f(x) + 4$
 - D) $h(x) = f(x) - 4$
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Question 3 [Easy]

If $f(x) = x^2$, which of the following represents the graph of $f(x)$ shifted 3 units left and 2 units down?

- A) $(x - 3)^2 - 2$
 - B) $(x + 3)^2 - 2$
 - C) $(x - 3)^2 + 2$
 - D) $(x + 3)^2 + 2$
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Question 4 [Easy]

The function $p(x) = -f(x)$. How does the graph of $y = p(x)$ compare to the graph of $y = f(x)$?

- A) Reflected over the x -axis
 - B) Reflected over the y -axis
 - C) Shifted down by 1 unit
 - D) Rotated 180 degrees about the origin
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Question 5 [Easy]

The vertex of a parabola $y = a(x - h)^2 + k$ is at (3, -7). After the parabola is shifted 5 units up, what is the new vertex?

- A) (3, -2)
 - B) (8, -7)
 - C) (3, -12)
 - D) (-2, -7)
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Question 6 [Medium]

The function q is defined by $q(x) = f(2x)$. If $f(6) = 10$, what is the value of $q(3)$?

- A) 5
 - B) 10
 - C) 20
 - D) 6
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Question 7 [Medium]

The graph of $y = f(x)$ has a maximum value of 9 at $x = 4$. What is the maximum value of $y = f(x) - 7$?

- A) 16
 - B) 9
 - C) 7
 - D) 2
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Question 8 [Medium]

If $g(x) = f(x - 3) + 2$, and the graph of f has an x -intercept at (5, 0), at what point does the corresponding feature appear on the graph of g ?

- A) (8, 2)
 - B) (2, 2)
 - C) (5, 2)
 - D) (8, 0)
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Question 9 [Medium]

The graph of $y = |x|$ is transformed to obtain $y = |x - 4| + 1$. Which of the following describes this transformation?

- A) Right 4, up 1
 - B) Left 4, up 1
 - C) Right 4, down 1
 - D) Left 4, down 1
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Question 10 [Medium]

The function $f(x) = (x - 1)^2 + 4$ is graphed in the xy -plane. The function g is defined by $g(x) = f(x + 6)$. What is the vertex of the graph of $y = g(x)$?

- A) $(-5, 4)$
 - B) $(7, 4)$
 - C) $(-5, 10)$
 - D) $(7, 10)$
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Question 11 [Medium]

If $h(x) = 3f(x)$, and the range of f is $[-2, 8]$, what is the range of h ?

- A) $[-2, 8]$
 - B) $[-6, 24]$
 - C) $[-6, 8]$
 - D) $[1, 11]$
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Question 12 [Medium]

The graph of $y = f(x)$ passes through the point $(-3, 5)$. If $g(x) = f(-x)$, through which point must the graph of $y = g(x)$ pass?

- A) $(3, 5)$
 - B) $(-3, -5)$
 - C) $(3, -5)$
 - D) $(-3, 5)$
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Question 13 [Medium]*Student-Produced Response*

The table shows selected values of a function f .

x :	-1	0	1	2	3
$f(x)$:	4	7	6	1	-2

What is the value of $f(x - 2)$ when $x = 3$?

Question 14 [Medium]

The minimum value of the function $f(x) = (x - 6)^2 + 3$ is 3. What is the minimum value of the function $g(x) = 2(x - 6)^2 + 3$?

- A) 1.5
 - B) 3
 - C) 6
 - D) 9
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Question 15 [Hard]

The graph of $y = f(x)$ is shown, where f has a domain of $[-4, 4]$. A new function g is defined by $g(x) = -f(x - 2) + 1$. What is the domain of g ?

- A) $[-2, 6]$
 - B) $[-6, 2]$
 - C) $[-4, 4]$
 - D) $[-3, 5]$
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Question 16 [Hard]***Student-Produced Response***

The function f is defined by $f(x) = 2x^2 - 8x + 11$. The graph of $y = f(x)$ in the xy -plane is a parabola. The function g is defined by $g(x) = f(x - 3)$. For what value of x does $g(x)$ reach its minimum?

Question 17 [Hard]

A function f has a y -intercept at $(0, -4)$. The function g is defined by $g(x) = f(3x) + 7$. What is the y -intercept of the graph of g ?

- A) $(0, 3)$
 - B) $(0, -12)$
 - C) $(0, 17)$
 - D) $(0, -4)$
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Question 18 [Hard]

Let $f(x) = x^3$. The graph of $g(x) = -(x + 2)^3 + 5$ can be obtained from the graph of f by applying which sequence of transformations?

- A) Left 2, reflect over x -axis, up 5
 - B) Right 2, reflect over y -axis, up 5
 - C) Left 2, reflect over y -axis, down 5
 - D) Right 2, reflect over x -axis, up 5
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Question 19 [Hard]***Student-Produced Response***

The function f is defined by $f(x) = |2x - 6| - 3$. For what value of x does $f(x)$ reach its minimum value?

Question 20 [Hard]

The function f has domain $[-5, 5]$ and range $[-3, 9]$. A new function g is defined by $g(x) = -f(2x) + 4$. What is the range of g ?

- A) $[-5, 7]$
 - B) $[-9, 3]$
 - C) $[-5, 3]$
 - D) $[-13, 7]$
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